

Module 02: Armstrong's Axioms & Inference Rules



$\rightarrow$ Roll_No\$.

2.2 Augmentation Rule (Additive Axiom)

Statement: Agar \$X

$\rightarrow$ Y\$, toh kisi bhi attribute set \$Z\$ ke liye \$XZ

$\rightarrow$ YZ\$ valid hoga.

Hindi: Dono taraf agar same attributes jod diye jayein, toh dependency intact rehti hai.

Example: Agar \$Roll_No

$\rightarrow$ Name\$, toh \$\{Roll_No, Semester\}

$\rightarrow$ \$\{Name, Semester\}\$.

2.3 Transitivity Rule (Chaining Axiom)

Statement: Agar \$X

$\rightarrow$ Y\$ aur \$Y

$\rightarrow$ Z\$, toh \$X

$\rightarrow$ Z\$ valid hoga.

Hindi: Dependency chain banakar aage propagate hoti hai.

Example: \$Roll_No

$\rightarrow$ Dept_ID\$ aur \$Dept_ID

$\rightarrow$ Dept_Name \$\implies\$ Roll_No

$\rightarrow$ Dept_Name\$.

3. Secondary / Derived Inference Rules (With Formal Proofs)

Primary axioms ko use karke hum practical calculations ke liye ye secondary rules derive karte hain:

3.1 Union / Additive Rule

Statement: Agar \$X

$\rightarrow$ Y\$ aur \$X

$\rightarrow$ Z\$, toh \$X

$\rightarrow$ YZ\$.

Formal Proof:

1. \$X

$\rightarrow$ Y\$ (Given)

2. X

$X \rightarrow X$ (Reflexivity) $\implies$ Augmenting step 1 by X : X

$X \rightarrow XY$.

3. X

$X \rightarrow Z$ (Given) $\implies$ Augmenting by Y : XY

$X \rightarrow YZ$.

4. By Transitivity (X

$X \rightarrow XY$ and X

$X \rightarrow YZ$): XY

$X \rightarrow YZ$. (Q.E.D.)

3.2 Decomposition / Projective Rule

Statement: Agar X

$X \rightarrow YZ$, toh X

$X \rightarrow Y$ aur X

$X \rightarrow Z$.

Formal Proof:

1. YZ

$X \rightarrow Y$ (Reflexivity, because $Y \subseteq YZ$).

2. X

$X \rightarrow YZ$ (Given).

3. By Transitivity: XY

$X \rightarrow Y$.

4. Similarly, YZ

$X \rightarrow Z$ (Reflexivity) $\implies$ By Transitivity: XY

$X \rightarrow Z$. (Q.E.D.)

Notice:

****CRITICAL AKTU EXAM TRAP:****

Right-Hand Side (RHS) attributes ko decompose kiya ja sakta hai (X

$X \rightarrow YZ \implies X$

$X \rightarrow Y, X$

$X \rightarrow Z$).

Par ****Left-Hand Side (LHS)** ko KABHI decompose nahi kiya ja sakta!**

$\$XY$

$\rightarrow Z \quad \text{or} \quad X$

$\rightarrow Z \quad \text{or} \quad Y$

$\rightarrow Z$

Example: $\$(Date_of_Birth, Mother_Name)$

$\rightarrow Person$, lekin akele $\$(Date_of_Birth$

or

$\rightarrow Person$!

$\rightarrow Person$!

3.3 Pseudo-Transitivity Rule

- **Statement:** Agar X

$\rightarrow Y$ aur WY

$\rightarrow Z$, toh WX

$\rightarrow Z$.

- **Formal Proof:**

1. X

$\rightarrow Y$ (Given) $\implies$ Augment by W : WX

$\rightarrow WY$.

2. WY

$\rightarrow Z$ (Given).

3. By Transitivity (WX

$\rightarrow WY$ and WY

$\rightarrow Z$): WX

$\rightarrow Z$. (Q.E.D.)

3.4 Composition Rule

- **Statement:** Agar X

$\rightarrow Y$ aur A

$\rightarrow B$, toh XA

$\rightarrow YB$.

- **Formal Proof:**

1. X

$\rightarrow Y \implies XA$

$\rightarrow Y A \$$ (Augment by $A \$$).

2. A

$\rightarrow B \implies Y A$

$\rightarrow Y B \$$ (Augment by $Y \$$).

3. By Transitivity: $A X A$

$\rightarrow Y B \** .

4. Soundness and Completeness (Theoretical Guarantees)

Armstrong's Axiom system computer science ke sabse robust axiomatic frameworks me se ek hai kyunki yeh do essential properties satisfy karta hai:

1. Soundness (Sabhi derived FDs sahi hain):

- Agar koi FD X

$\rightarrow Y \$$ Armstrong axioms se derive hoti hai, toh wo relation $R \$$ ke har legal instance me 100% physically valid hogi. Koi bhi wrong/false dependency derive nahi ho sakti!

2. Completeness (Koi bhi valid FD chootegi nahi):

- Agar koi FD X

$\rightarrow Y \$$ relation schema ke rules ke hisaab se logically imply hoti hai, toh wo Armstrong axioms ke finite steps se zaroor derive ho sakti hai. Iska matlab closure $F^+ \$$ poori tarah calculate ho sakta hai!

5. Architectural Diagram

6. Solved AKTU Practice Derivation

Problem: Given $F = \{A$

$\rightarrow B, B$

$\rightarrow C, CD$

$\rightarrow E \setminus \$$. Prove that AD

$\rightarrow E \$$ holds using Armstrong's Axioms.

Solution Step-by-Step:

1. A

$\rightarrow B \$$ (Given)

2. B

$\rightarrow C \$$ (Given)

3. From (1) and (2), by **Transitivity Rule**: A

$\Rightarrow C$.

4. Take A

$\Rightarrow C$ and augment both sides with attribute D (**Augmentation Rule**):

AD

$\Rightarrow CD$

5. Given CD

$\Rightarrow E$.

6. From step 4 (AD

$\Rightarrow CD$) and step 5 (CD

$\Rightarrow E$), by **Transitivity Rule**:

$\mathbf{AD}$

$\Rightarrow E$

Hence Proved!